K. M. Mynurdoza

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Research Interests

- **Multiphysics Simulation & Continuum Mechanics:** Finite element analysis (FEA), conjugate heat transfer, computational fluid dynamics.
- **Robotic Systems & Mechatronics:** Mechanical mechanism design, precision motion transmission, real-time embedded control and advanced control.

Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh

Bachelor of Science in Mechanical Engineering *Nov 2023 – Expected Jul 2028*

- Current Standing: Level-3, Term-I (Completed 80.00 credit hours across 4 semesters).
- Cumulative Grade Point Average: **3.70 / 4.00** (Recent Term GPA: **3.83 / 4.00** in Level-2 Term-II).

Sylhet Cadet College Sylhet, Bangladesh

Higher Secondary Certificate (HSC) – Science *Passing Year: 2022*

- **GPA: 5.00 / 5.00** · Awarded Government Talentpool Scholarship.

Sylhet Cadet College Sylhet, Bangladesh

Secondary School Certificate (SSC) – Science *Passing Year: 2020*

- **GPA: 5.00 / 5.00** · Awarded Government Talentpool Scholarship · Ranked **7th** in Sylhet Education Board.

Research & Engineering Projects

Finite Element Thermal Analysis of Oil-Cooled DC-Link Capacitors *Ongoing Research*

- Investigating conjugate heat transfer and thermal-fluid behavior in dielectric oil-cooled DC-link film capacitors for high-power converter applications.
- Formulating finite element models to simulate internal temperature fields, fluid flow velocity distributions, and localized hotspot generation.
- Analyzing forced convective cooling pathways and fluid circulation geometries to mitigate thermal stress concentrations and improve operational lifespan.

Thermomechanical Simulation of Power Module Solder Joints (COMSOL) *FEA / Reliability*

- Replicated failure mechanisms and viscoplastic creep in the solder layer of a Single-Sided Cooling (SSC) power module (Semikron SKM50GB123D) using 2D half-symmetry coupled thermal-structural analysis.
- Implemented the nonlinear **Anand Viscoplasticity Model** for 60Sn40Pb solder to capture strain-rate- and temperature-dependent inelastic deformation under cyclic thermal loading ($\Delta T_j \approx 120°C$).
- Utilized segregated solvers with strict time stepping and targeted solder corner mesh refinement; evaluated Von Mises stress (150–160 MPa) and creep dissipation energy (ΔW_{sc}) with the **Morrow fatigue model** for lifetime prediction.

"Beyond Horizon" — Automated Alt-Azimuth Robotic Mount Code & Design Repository

- Designed and constructed a dual-axis motorized alt-azimuth mount in SolidWorks for celestial and terrestrial observation.
- Engineered a custom reduction gear train driven by NEMA stepper motors to achieve smooth, vibration-isolated angular motion.
- Developed embedded firmware in C++ that performs real-time coordinate transformations from equatorial (RA/Dec) to local topocentric (Alt/Az) coordinates for automated pointing.
- Conducted structural finite element analysis of the objective doublet lens cell (D80F900) in **COMSOL Multiphysics** to evaluate chromatic aberration and optical performance.

Honors & Awards

- **Award of Merit** – *6th International Blockchain Olympiad (IBCOL)*, Amsterdam, Netherlands, 2023.
- **National Round Winner** – *Bangladesh Physics Olympiad (BdPhO)*, 2022.
- **Divisional & Selection Winner** – *Bangladesh Mathematical Olympiad (BdMO)*, 2022 & 2023.

Technical Skills

- **Simulation & Multiphysics:** COMSOL Multiphysics, ANSYS (Fluent, Mechanical), Finite Element Analysis (FEA), Conjugate Heat Transfer.
- **CAD & Prototyping:** SolidWorks (Parametric 3D Modeling, Assemblies, Mechanism Synthesis), AutoCAD, 3D Printing, Precision Machining.
- **Embedded Systems & Robotics:** Stepper Motor Actuation, Sensor Interfacing, Arduino, STM32, C/C++ Firmware, ROS.
- **Programming & Scientific Tools:** Python , C, C++, MATLAB, LaTeX.

Leadership, Extracurriculars & Affiliations

- **Student Affiliate Member, Institution of Mechanical Engineers (IMechE):** Active in technical seminars and BUET student chapter initiatives.
- **Chess.com College Ambassador (School Year 2026–2027):** Selected as official campus ambassador representing BUET; organizing collegiate chess events, community outreach, and promoting student competitive engagement.
- **Treasurer, BUET Chess Club (Jul 2026 – Present):** Managing club accounts, operational budgeting, and tournament logistics for university-wide and national inter-collegiate championships.
- **FIDE-Rated Chess Player:** Officially rated tournament player under the International Chess Federation (FIDE); actively competing in national rating events and collegiate chess championships.